3 Fig. 3.1 shows a boy throwing a ball at an object in a fairground.

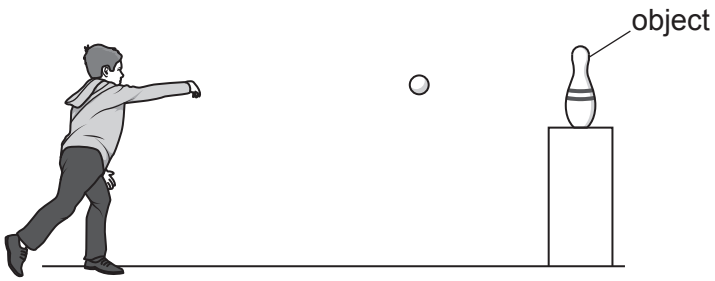


Fig. 3.1

The ball has a mass of 190 g and travels horizontally with a constant speed of 6.9 m / s.

(a) Calculate the momentum of the ball.

momentum = ... [2]

(b) After hitting the object, the ball bounces back along the same straight path with a speed of 1.5 m / s. The object has a mass of 1.8 kg.

Calculate the speed of the object after it is hit by the ball.

speed = ... [3]

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(c) The kinetic energy of the ball is 4.5 J before the collision and 0.2 J after the collision.

Calculate the change in total kinetic energy of the ball and object during the collision.

change in total kinetic energy = ... [3]

[Total: 8]